

A Gentle Tutorial on Multiplication in $GF(2^8)$

Why this tutorial exists

This is a personal learning guide for understanding how multiplication works in $GF(2^8)$, especially the kind used in cryptography, such as AES. The goal is not to rush into formal abstract algebra. The goal is to make the machinery visible.

The main example is:

$$87 \cdot 131 = 193$$

inside the AES version of $GF(2^8)$.

In hexadecimal, this is:

$$0x57 \cdot 0x83 = 0xC1$$

This does **not** mean ordinary decimal multiplication. Ordinary multiplication gives:

$$87 \cdot 131 = 11397$$

But Galois field multiplication is a different kind of multiplication. It treats bytes as tiny polynomials and then folds the answer back into one byte.

Think of it like building with square tiles on a board where the board has a special wraparound rule. You can temporarily spill beyond the board, but then the field rules tell you exactly how to fold the extra pieces back into the allowed space.

Part 1: The Big Idea

What is $GF(2^8)$?

$GF(2^8)$ is a finite field with 256 possible values.

That means its values fit exactly in one byte:

00000000 through 11111111

or decimal:

0 through 255

or hexadecimal:

$0x00$ through $0xFF$

In normal byte arithmetic, numbers can overflow. But in $GF(2^8)$, we do not just throw away overflow. We reduce it using a special polynomial rule.

That special rule is what makes the math useful for cryptography.

Part 2: Bytes as Polynomials

In $GF(2^8)$, a byte can be read as a polynomial.

The eight bit positions correspond to powers of x :

Bit position	7	6	5	4	3	2	1	0
Polynomial power	x^7	x^6	x^5	x^4	x^3	x^2	x	1

A bit value of 1 means that power is present. A bit value of 0 means it is absent.

Example: 87

First convert 87 to binary:

$$87 = 01010111_2$$

Now map the 1 bits to powers of x :

Polynomial power	x^7	x^6	x^5	x^4	x^3	x^2	x	1
Bit in 87	0	1	0	1	0	1	1	1

So:

$$87 = x^6 + x^4 + x^2 + x + 1$$

Example: 131

$$131 = 10000011_2$$

So:

$$131 = x^7 + x + 1$$

So our original problem becomes:

$$(x^6 + x^4 + x^2 + x + 1)(x^7 + x + 1)$$

This is the first major shift:

In $GF(2^8)$, we do not initially think of 87 and 131 as ordinary numbers. We think of them as polynomial shapes made out of bits.

Part 3: Addition Means XOR

This is one of the most important rules.

In $GF(2)$, coefficients are only 0 or 1.

So addition works like XOR:

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0$$

That last one is the strange one if you are coming from ordinary arithmetic.

It means pairs cancel.

So:

$$x^7 + x^7 = 0$$

$$x^2 + x^2 = 0$$

$$x + x = 0$$

A helpful image: imagine each term is a light switch. If it appears once, it turns on. If it appears twice, it toggles back off. The polynomial that remains is whatever lights are still on.

Part 4: Multiply First, Then Reduce

The multiplication has two phases:

1. **Polynomial multiplication:** multiply normally using powers of x , but combine terms using XOR.

2. **Reduction:** fold any powers x^8 or higher back into one byte using the irreducible polynomial.

So the workflow is:

bytes $\rightarrow$ polynomials $\rightarrow$ multiply $\rightarrow$ cancel pairs $\rightarrow$ reduce $\rightarrow$ byte

Part 5: The AES Irreducible Polynomial

AES uses this polynomial:

$$m(x) = x^8 + x^4 + x^3 + x + 1$$

This is often represented in hexadecimal as:

$0x11B$

In binary:

1 0001 1011

That is a 9-bit pattern because it includes the x^8 term.

What does this polynomial do?

It is the wraparound rule.

In this field, we treat:

$$x^8 + x^4 + x^3 + x + 1 = 0$$

Therefore:

$$x^8 = x^4 + x^3 + x + 1$$

That means whenever x^8 appears, we can replace it with:

$$x^4 + x^3 + x + 1$$

This is the heart of reduction.

Analogy: modular arithmetic

In ordinary modular arithmetic, if we are working mod 10, then:

$$13 \equiv 3 \pmod{10}$$

because 10 wraps around.

In $GF(2^8)$, we are doing a similar idea, except the thing doing the wrapping is a polynomial instead of an ordinary number.

So instead of:

$$10 \equiv 0$$

we have:

$$x^8 + x^4 + x^3 + x + 1 \equiv 0$$

This is why the irreducible polynomial feels odd at first. It is not just a number. It is the rule that defines how high-degree polynomial terms get folded back into the 8-bit world.

Why “irreducible”?

For now, the simplest explanation is:

An irreducible polynomial acts like a prime number does in ordinary modular arithmetic.

When you do arithmetic modulo a prime number, every nonzero number has a multiplicative inverse.

Likewise, when you do polynomial arithmetic modulo an irreducible polynomial, every nonzero field element has a multiplicative inverse.

That is why it creates a proper field rather than a broken arithmetic system.

But at the beginning, you can safely remember:

The irreducible polynomial is the official folding rule of the field.

Part 6: Full Example: $87 \cdot 131$ in AES $GF(2^8)$

Step 1: Convert both numbers to binary

$$87 = 01010111_2$$

$$131 = 10000011_2$$

Step 2: Convert both bytes to polynomials

$$01010111_2 = x^6 + x^4 + x^2 + x + 1$$

$$10000011_2 = x^7 + x + 1$$

So:

$$87 \cdot 131$$

becomes:

$$(x^6 + x^4 + x^2 + x + 1)(x^7 + x + 1)$$

Step 3: Distribute each term

Multiply each term from the first polynomial by the whole second polynomial.

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\begin{array}{c|c} \text{term from } 87 & \text{result} \\ \hline x^6 & x^{13}+x^7+x^6 \\ x^4 & x^{11}+x^5+x^4 \\ x^2 & x^9+x^3+x^2 \\ x & x^8+x^2+x \\ 1 & x^7+x+1 \end{array}
```

Now collect everything:

$$x^{13} + x^7 + x^6 + x^{11} + x^5 + x^4 + x^9 + x^3 + x^2 + x^8 + x^2 + x + x^7 + x + 1$$

Reorder by degree:

$$x^{13} + x^{11} + x^9 + x^8 + x^7 + x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x^2 + x + x + 1$$

Step 4: Cancel duplicate pairs

Because addition is XOR:

$$x^7 + x^7 = 0$$

$$x^2 + x^2 = 0$$

$$x + x = 0$$

So the raw product becomes:

$$x^{13} + x^{11} + x^9 + x^8 + x^6 + x^5 + x^4 + x^3 + 1$$

This is the product before reduction.

But it is too large. It contains powers above x^7 , so it does not fit in one byte yet.

Step 5: Reduce the highest term, x^{13}

The AES reduction polynomial is:

$$x^8 + x^4 + x^3 + x + 1$$

We want to cancel x^{13} .

Since:

$$x^8 \cdot x^5 = x^{13}$$

multiply the AES polynomial by x^5 :

$$x^5(x^8 + x^4 + x^3 + x + 1)$$

Distribute:

$$x^{13} + x^9 + x^8 + x^6 + x^5$$

Now XOR this with the raw product:

$$x^{13} + x^{11} + x^9 + x^8 + x^6 + x^5 + x^4 + x^3 + 1 \oplus x^{13} + x^9 + x^8 + x^6 + x^5$$

Cancel matching terms:

- x^{13} cancels
- x^9 cancels
- x^8 cancels
- x^6 cancels
- x^5 cancels

Left over:

$$x^{11} + x^4 + x^3 + 1$$

Still too large because of x^{11} .

Step 6: Reduce x^{11}

Again use:

$$x^8 + x^4 + x^3 + x + 1$$

To cancel x^{11} , multiply by x^3 :

$$x^3(x^8 + x^4 + x^3 + x + 1)$$

Distribute:

$$x^{11} + x^7 + x^6 + x^4 + x^3$$

Now XOR with what remained:

$$x^{11} + x^4 + x^3 + 1 \oplus x^{11} + x^7 + x^6 + x^4 + x^3$$

Cancel matching terms:

- x^{11} cancels
- x^4 cancels
- x^3 cancels

Left over:

$$x^7 + x^6 + 1$$

Now we are finished, because the largest power is x^7 . It fits into one byte.

Step 7: Convert back to binary

$$x^7 + x^6 + 1$$

means bits 7, 6, and 0 are set:

Polynomial power	x^7	x^6	x^5	x^4	x^3	x^2	x	1
Bit in result	1	1	0	0	0	0	0	1

So:

$$x^7 + x^6 + 1 = 11000001_2$$

$$11000001_2 = 0xC1 = 193$$

Therefore:

$87 \cdot 131 = 193 \text{ in AES } GF(2^8)$

or:

$0x57 \cdot 0x83 = 0xC1$

Part 7: A Smaller Toy Example in $GF(2^4)$

Before doing more $GF(2^8)$, it helps to shrink the playground.

Let's use $GF(2^4)$, where values fit in 4 bits instead of 8.

The powers allowed are:

$$x^3, x^2, x, 1$$

We need a degree-4 irreducible polynomial. A common one is:

$$x^4 + x + 1$$

That gives the wraparound rule:

$$x^4 = x + 1$$

Now let's multiply two 4-bit values.

Example: $0b1011 \cdot 0b0110$

In decimal:

$$0b1011 = 11$$

$$0b0110 = 6$$

But again, this is not ordinary multiplication.

Convert to polynomials:

$$1011_2 = x^3 + x + 1$$

$$0110_2 = x^2 + x$$

Multiply:

$$(x^3 + x + 1)(x^2 + x)$$

Distribute:

$$x^3(x^2 + x) = x^5 + x^4$$

$$x(x^2 + x) = x^3 + x^2$$

$$1(x^2 + x) = x^2 + x$$

Collect:

$$x^5 + x^4 + x^3 + x^2 + x^2 + x$$

Cancel duplicate x^2 :

$$x^5 + x^4 + x^3 + x$$

Now reduce using:

$$x^4 = x + 1$$

First handle x^5 :

$$x^5 = x(x^4) = x(x + 1) = x^2 + x$$

So:

$$x^5 + x^4 + x^3 + x$$

becomes:

$$(x^2 + x) + (x + 1) + x^3 + x$$

Collect:

$$x^3 + x^2 + x + x + x + 1$$

Since three x terms appear, two cancel and one remains:

$$x^3 + x^2 + x + 1$$

So the result is:

$$1111_2$$

Therefore:

$$0b1011 \cdot 0b0110 = 0b1111$$

in this particular $GF(2^4)$ field.

Why this small example helps

The $GF(2^4)$ version is the same dance as $GF(2^8)$, just with fewer powers.

In $GF(2^4)$, the fold starts at x^4 .

In $GF(2^8)$, the fold starts at x^8 .

Same idea. Bigger room.

Part 8: Another $AES\ GF(2^8)$ Example

Let's multiply:

$$0x03 \cdot 0x57$$

This is a useful example because $0x03$ is simple.

Convert:

$$0x03 = 00000011_2 = x + 1$$

$$0x57 = 01010111_2 = x^6 + x^4 + x^2 + x + 1$$

Now multiply:

$$(x + 1)(x^6 + x^4 + x^2 + x + 1)$$

Distribute:

$$x(x^6 + x^4 + x^2 + x + 1) = x^7 + x^5 + x^3 + x^2 + x$$

$$1(x^6 + x^4 + x^2 + x + 1) = x^6 + x^4 + x^2 + x + 1$$

Collect:

$$x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x^2 + x + x + 1$$

Cancel duplicates:

$$x^2 + x^2 = 0$$

$$x + x = 0$$

Left over:

$$x^7 + x^6 + x^5 + x^4 + x^3 + 1$$

This already fits within x^7 , so no reduction is needed.

Convert back to bits:

$$11111001_2 = 0xF9$$

So:

$$0x03 \cdot 0x57 = 0xF9$$

Notice something nice:

Not every multiplication requires reduction. Reduction only happens when powers x^8 or higher appear.

Part 9: Another AES Example That Needs Reduction

Now try:

$$0x02 \cdot 0x57$$

This is equivalent to multiplying by x .

$$0x02 = 00000010_2 = x$$

$$0x57 = x^6 + x^4 + x^2 + x + 1$$

Multiply:

$$\begin{aligned} x(x^6 + x^4 + x^2 + x + 1) \\ = x^7 + x^5 + x^3 + x^2 + x \end{aligned}$$

This fits in one byte.

So:

$$x^7 + x^5 + x^3 + x^2 + x = 10101110_2 = 0xAE$$

Therefore:

$$0x02 \cdot 0x57 = 0xAE$$

Now multiply $0x02 \cdot 0xAE$.

$$0xAE = 10101110_2 = x^7 + x^5 + x^3 + x^2 + x$$

Multiply by x :

$$x^8 + x^6 + x^4 + x^3 + x^2$$

Now x^8 appears, so reduce:

$$x^8 = x^4 + x^3 + x + 1$$

Substitute:

$$(x^4 + x^3 + x + 1) + x^6 + x^4 + x^3 + x^2$$

Cancel duplicate terms:

$$x^4 + x^4 = 0$$

$$x^3 + x^3 = 0$$

Left over:

$$x^6 + x^2 + x + 1$$

Convert to bits:

$$01000111_2 = 0x47$$

So:

$$0x02 \cdot 0xAE = 0x47$$

This is exactly the same idea as the common AES operation called `xtime`.

Part 10: Why AES Code Often XORs with 0x1B

The AES polynomial is:

$$x^8 + x^4 + x^3 + x + 1$$

The reduction rule is:

$$x^8 = x^4 + x^3 + x + 1$$

The right side is:

$$x^4 + x^3 + x + 1$$

As an 8-bit pattern, that is:

$$00011011_2$$

which is:

$$0x1B$$

That is why AES implementations often do this when multiplying by x :

1. Left shift the byte.
2. If the original top bit was 1, XOR with $0x1B$.

Why?

Because a top bit of 1 means an x^8 term spilled out after the shift.

XORing with $0x1B$ performs the fold:

$$x^8 \mapsto x^4 + x^3 + x + 1$$

Analogy: the conveyor belt

Imagine an 8-slot conveyor belt labeled:

$$x^7, x^6, x^5, x^4, x^3, x^2, x, 1$$

Multiplying by x shifts every object one slot to the left.

If nothing falls off the left edge, you are done.

If something falls off the left edge, that fallen x^8 tile is not discarded. It gets broken into four replacement tiles:

$$x^4, x^3, x, 1$$

XORing with $0x1B$ places those replacement tiles onto the conveyor.

Part 11: Summary of the Whole Method

To multiply two bytes in AES $GF(2^8)$:

1. Convert each byte to binary.
2. Interpret the 1 bits as powers of x .
3. Multiply the polynomials.
4. Combine like terms using XOR, so duplicate pairs cancel.
5. If any powers are x^8 or higher, reduce using:

$$x^8 + x^4 + x^3 + x + 1$$

1. Convert the final polynomial back to an 8-bit value.

The most compact mental model:

Bits are polynomial coefficients. Addition is XOR. Multiplication is polynomial multiplication.
The irreducible polynomial is the fold-back rule.

Part 12: Practice Problems

Practice 1

Convert this byte to a polynomial:

$$10100101_2$$

Answer:

$$x^7 + x^5 + x^2 + 1$$

Practice 2

Convert this polynomial to a byte:

$$x^6 + x^3 + x + 1$$

Answer:

$$01001011_2 = 0x4B$$

Practice 3

Multiply in polynomial form before reduction:

$$(x^3 + x + 1)(x^2 + 1)$$

Distribute:

$$x^3(x^2 + 1) = x^5 + x^3$$

$$x(x^2 + 1) = x^3 + x$$

$$1(x^2 + 1) = x^2 + 1$$

Collect:

$$x^5 + x^3 + x^3 + x + x^2 + 1$$

Cancel duplicate x^3 :

$$x^5 + x^2 + x + 1$$

Answer before reduction:

$$x^5 + x^2 + x + 1$$

Practice 4

In AES $GF(2^8)$, reduce:

$$x^8 + x^6 + x^2$$

Use:

$$x^8 = x^4 + x^3 + x + 1$$

Substitute:

$$(x^4 + x^3 + x + 1) + x^6 + x^2$$

Answer:

$$x^6 + x^4 + x^3 + x^2 + x + 1$$

Binary:

$$01011111_2 = 0x5F$$

Part 13: What to Remember When It Feels Foggy

If this feels slippery, that is normal. The reason is that several familiar words are wearing unfamiliar costumes:

- “Number” means byte interpreted as a polynomial.
- “Addition” means XOR.
- “Multiplication” means polynomial multiplication.
- “Modulo” means reducing by a polynomial, not an integer.
- “Irreducible” means prime-like.

The central image is this:

A byte is an 8-slot polynomial machine. Multiplication can create terms that spill beyond the machine. The irreducible polynomial tells you how to fold those terms back into the machine without losing the algebraic structure.

That is the beginning of the mathematics behind AES-style finite field arithmetic.

Part 14: The Worked Example in One Page

$$87 = 01010111_2 = x^6 + x^4 + x^2 + x + 1$$

$$131 = 10000011_2 = x^7 + x + 1$$

Multiply:

$$(x^6 + x^4 + x^2 + x + 1)(x^7 + x + 1)$$

Rows:

$$x^6 \mid x^{13} + x^7 + x^6 \quad x^4 \quad x^{11} + x^5 + x^4 \quad x^2 \quad x^9 + x^3 + x^2 \quad x \quad x^8 + x^2 + x \quad 1 \quad x^7 + x + 1$$

Collect and cancel pairs:

$$x^{13} + x^{11} + x^9 + x^8 + x^6 + x^5 + x^4 + x^3 + 1$$

Reduce x^{13} :

$$x^5(x^8 + x^4 + x^3 + x + 1) = x^{13} + x^9 + x^8 + x^6 + x^5$$

XOR leaves:

$$x^{11} + x^4 + x^3 + 1$$

Reduce x^{11} :

$$x^3(x^8 + x^4 + x^3 + x + 1) = x^{11} + x^7 + x^6 + x^4 + x^3$$

XOR leaves:

$$x^7 + x^6 + 1$$

Convert back:

$$x^7 + x^6 + 1 = 11000001_2 = 0xC1 = 193$$

Final answer:

$0x57 \cdot 0x83 = 0xC1$

$$87 \cdot 131 = 193 \text{ in AES } GF(2^8)$$